



Lab 6: More on GitHub Actions

Files you will create (local)

Create a folder **scripts/** and the following files:

math_utils.py

```
def add(a, b):  
    return a + b  
  
def subtract(a, b):  
    return a - b
```

test_math_utils.py

```
from math_utils import add, subtract  
  
def test_add():  
    assert add(2, 3) == 5  
  
def test_subtract():  
    assert subtract(5, 3) == 2
```

Commit these files first. (You can use the previous lab's repo or create a new one)

git add .

git commit -m "Add math utils and pytest tests"

git push origin main



Exercise 1: Run Automated Tests

Create a workflow that installs pytest and runs the tests on every push and pull_request.

1. In your repo create the directory if it does not exist: **.github/workflows**
2. Inside it create file **pytest.yml** with exact content:

```
name: Run Automated Tests

on: [push, pull_request]

jobs:
  test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: '3.10'

      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install pytest

      - name: Run pytest
        run: pytest -v scripts/
```

3. Add, commit, push
4. Check the output in your repo on Actions tab.



Exercise 2: Using Matrix Build Across Operating Systems

Objective: Run the same pytest suite on Ubuntu, Windows, and macOS using a matrix strategy.

1. Replace **runs-on: ubuntu-latest** line in the last exercise with:
Make sure you fix any indentation errors.

```
strategy:
  matrix:
    os: [ubuntu-latest, windows-latest, macos-latest]
  runs-on: ${ matrix.os }
```

2. Add, commit, push
3. Check the output in your repo on Actions tab.



Exercise 3: Uploading an Artifact

Generate a file inside the workflow and upload it as an artifact so you can download it after the workflow finishes.

1. Replace the last step in the previous exercise with this content:

```
- name: Run pytest (generate junit xml)
  run: pytest -v scripts/ --junitxml=pytest-report.xml

- name: Upload test report artifact
  uses: actions/upload-artifact@v4
  with:
    name: pytest-report_${{ matrix.os }}
    path: pytest-report.xml
```

2. Add, commit, push
3. Check the output in your repo on Actions tab.

You can find the uploaded artifacts under the job execution graph.

Click `pytest-report_XXX` → download `pytest-report.xml`



Deliverables

A pdf document containing screenshots of:

Exercise 1:

- The Actions tab run page for Run Pytest showing tests passed.

Exercise 2:

- The Actions tab showing the matrix run with three jobs (Ubuntu / Windows / macOS).

Exercise 3:

- A screenshot of the downloaded pytest-report.xml